

Experiment No. 6

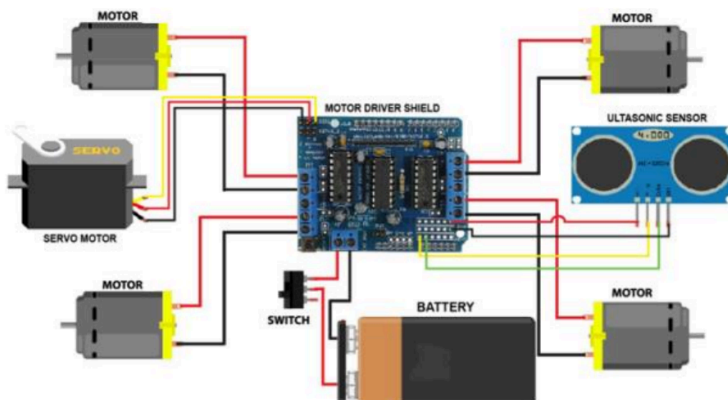
AIM: Interfacing & Simulation with Arduino (Obstacle Avoiding Car)

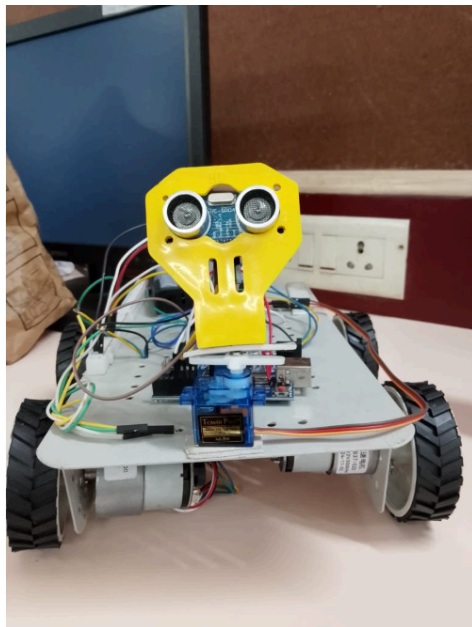
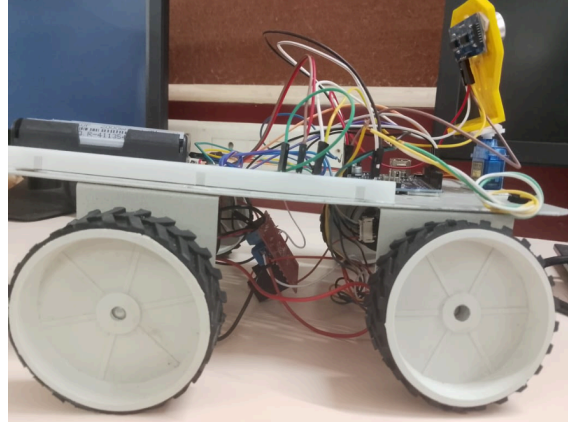
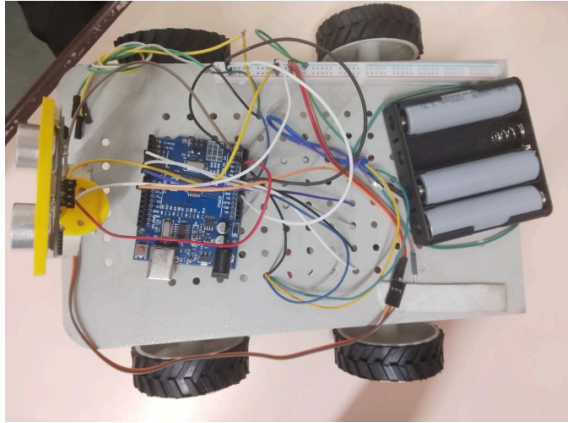
Introduction: An obstacle-avoiding car is an autonomous robotic vehicle that uses ultrasonic sensors to detect obstacles in its path and steer itself accordingly. The purpose of this experiment is to build a simple obstacle-avoiding car using an Arduino board, motor driver shield, ultrasonic sensor, DC motors, and a servo motor.

Components Required:

- **Arduino Uno:** Acts as the brain of the robot, processing sensor data and controlling the motors.
- **Motor Driver Shield:** Sits on top of the Arduino Uno and provides connections for multiple motors. It simplifies controlling motor speed and direction through PWM signals.
- **Ultrasonic Sensor (HC-SR04):** Measures the distance to obstacles using sound waves.
- **Servo Motor:** Used to rotate the ultrasonic sensor for scanning different directions (left, right, and straight).
- **DC Motors (4 motors):** Drive the car forward, backward, left, or right.
- **Battery:** Supplies power to the motors and Arduino.
- **Switch:** Used to turn the system on or off.
- **Connecting Wires:** For establishing connections between components.

Connections:





1. **Motor Driver Shield:** Connected to Arduino Uno for controlling four DC motors using PWM signals. Powered by external battery, PWM pins (ENA, ENB) connected to Arduino PWM pins (e.g., 9, 10), and motor control pins connected to digital pins (e.g., 3, 4, 5, 6).
2. **Ultrasonic Sensor (HC-SR04):**
 - VCC → 5V (Arduino)
 - GND → GND (Arduino)
 - TRIG → Digital pin 9 (Arduino) (Sends ultrasonic waves)
 - ECHO → Digital pin 10 (Arduino) (Receives reflected waves)
3. **Servo Motor:** Connected to the motor driver shield to rotate and scan the environment.

4. **DC Motors:** Connected to the motor driver shield. Two motors on the left and two on the right for differential steering.
5. **Battery & Switch:** The battery powers the motors through the motor driver shield, and the switch helps in turning the system on/off.

Working Principle: The ultrasonic sensor continuously sends out sound waves. When these waves hit an obstacle, they bounce back to the sensor. The sensor calculates the distance to the obstacle by measuring the time taken for the waves to return. If the distance is below a certain threshold (say, 20 cm), the car takes evasive action. The servo motor rotates the ultrasonic sensor left and right to scan for a clear path. The Arduino processes this data and controls the motors to steer the car away from obstacles.

Software Implementation:

1. **Library Inclusion:** Includes libraries for motor control and ultrasonic sensor handling.
2. **Initialization:** Sets up motor pins, ultrasonic sensor pins, and servo motor.
3. **Distance Measurement:** Continuously checks distance from obstacles using the ultrasonic sensor.
4. **Decision Making:** If an obstacle is detected within a threshold, the car stops. The servo motor rotates the sensor left and right to determine a clear path. The car then turns in the direction with the most clearance and continues moving.
5. **Continuous Operation:** Repeats the above steps to maintain obstacle avoidance.

Communication:

- **PWM Signals:** Control the speed of the motors.
- **Digital Signals:** Used for triggering the ultrasonic sensor and reading the echo.
- **Servo Motor Control:** Uses PWM for rotating the ultrasonic sensor for scanning.

Conclusion: The experiment successfully demonstrates the use of an ultrasonic sensor, motor driver shield, and servo motor controlled by an Arduino to create an obstacle-avoiding car. This project showcases practical concepts of distance measurement, motor control, and decision-making algorithms.